DOCKER VOLUME AND .

1. **Docker Volume:**

What it really means:

- Docker manages the storage
- Data survives container deletion

Pros:

- Fast
- Safe
- No permission drama
- Best for **production**

Cons:

- Not human-readable on host
- Not for live editing code

```
bash: ta: command not found
[root@ip-172-31-30-82 ec2-user]# docker run -itd -p 8080:80 --name newcontainer --mount type=volume,source=mydata,target=/app/data httpd
f8309383c7670e30b3e1dc95db96c34b67f9ecca31ff5189c3066d4d0bb9dab2
[root@ip-172-31-30-82 ec2-user]# docker ps
CONTAINER ID   IMAGE     COMMAND              CREATED         STATUS         PORTS                                   NAMES
f8309383c767   httpd     "httpd-foreground"   8 seconds ago   Up 8 seconds   0.0.0.0:8080->80/tcp, :::8080->80/tcp   newcontainer
[root@ip-172-31-30-82 ec2-user]# docker exec -it newcontainer
"docker exec" requires at least 2 arguments.
See 'docker exec --help'.

Usage:  docker exec [OPTIONS] CONTAINER COMMAND [ARG...]

Execute a command in a running container
[root@ip-172-31-30-82 ec2-user]# docker exec -it newcontainer /bin/bash
root@f8309383c767:/usr/local/apache2# cd /var/lib/docker/volumes
bash: cd: /var/lib/docker/volumes: No such file or directory
root@f8309383c767:/usr/local/apache2# cd /var/lib/docker/volume
```

```
root@f8309383c767:/usr/local/apache2# cd/var/lib/docker/volumes
bash: cd/var/lib/docker/volumes: No such file or directory
root@f8309383c767:/usr/local/apache2# ls
bin  build  cgi-bin  conf  error  htdocs  icons  include  logs  modules
root@f8309383c767:/usr/local/apache2# /app/data
bash: /app/data: Is a directory
root@f8309383c767:/usr/local/apache2# cd/var/lib/docker/volume
bash: cd/var/lib/docker/volume: No such file or directory
root@f8309383c767:/usr/local/apache2# touch filenew
root@f8309383c767:/usr/local/apache2# cd/var/lib/docker/volume
bash: cd/var/lib/docker/volume: No such file or directory
root@f8309383c767:/usr/local/apache2# ls
bin  build  cgi-bin  conf  error  filenew  htdocs  icons  include  logs  modules
root@f8309383c767:/usr/local/apache2# cd /app/data
root@f8309383c767:/app/data# ls
root@f8309383c767:/app/data# touch file1
root@f8309383c767:/app/data# ls
file1
root@f8309383c767:/app/data# exit
exit
[root@ip-172-31-30-82 ec2-user]# cd /var/lib/docker/volumes
```

i-0afc3df3d18c02946 (dockervolume)

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```
root@f8309383c767:/usr/local/apache2# ls
bin  build  cgi-bin  conf  error  htdocs  icons  include  logs  modules
root@f8309383c767:/usr/local/apache2# /app/data
bash: /app/data: Is a directory
root@f8309383c767:/usr/local/apache2# cd/var/lib/docker/volume
bash: cd/var/lib/docker/volume: No such file or directory
root@f8309383c767:/usr/local/apache2# touch filenew
root@f8309383c767:/usr/local/apache2# cd/var/lib/docker/volume
bash: cd/var/lib/docker/volume: No such file or directory
root@f8309383c767:/usr/local/apache2# ls
bin  build  cgi-bin  conf  error  filenew  htdocs  icons  include  logs  modules
root@f8309383c767:/usr/local/apache2# cd /app/data
root@f8309383c767:/app/data# ls
root@f8309383c767:/app/data# touch file1
root@f8309383c767:/app/data# ls
file1
root@f8309383c767:/app/data# exit
exit
[root@ip-172-31-30-82 ec2-user]# cd /var/lib/docker/volumes
[root@ip-172-31-30-82 volumes]# ls
backingfsblockdev  metadata.db  mydata
[root@ip-172-31-30-82 volumes]# cd mydata
[root@ip-172-31-30-82 mydata]# ls
_data
[root@ip-172-31-30-82 mydata]# cd _data
[root@ip-172-31-30-82 _data]# ls
file1
[root@ip-172-31-30-82 _data]#
```

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The docker volume commands are:

```
[root@ip-172-31-24-189 _data]# docker volume

Usage:  docker volume COMMAND

Manage volumes

Commands:
  create      Create a volume
  inspect     Display detailed information on one or more volumes
  ls          List volumes
  prune       Remove unused local volumes
  rm          Remove one or more volumes
```

2. Docker Bind Mount

What it really means

- Your **host folder** is mounted directly into the container
- Changes on host = changes in container (live sync)

Pros

- Perfect for **development**
- Hot reload with Node.js

Cons

- OS permission issues (Linux/Windows pain)
- Slower I/O on Windows/Mac
- You can accidentally overwrite container files

```
[root@ip-172-31-30-82 ec2-user]# sudo su
[root@ip-172-31-30-82 ec2-user]# exit
exit
[root@ip-172-31-30-82 ec2-user]# cd
[root@ip-172-31-30-82 ~]# pwd
/root
[root@ip-172-31-30-82 ~]# exit
exit
[ec2-user@ip-172-31-30-82 ~]$ mkdir /temp/mydata
mkdir: cannot create directory '/temp/mydata': No such file or directory
[ec2-user@ip-172-31-30-82 ~]$ sudo su
[root@ip-172-31-30-82 ec2-user]# mkdir /tmp/mydata
[root@ip-172-31-30-82 ec2-user]#  docker run -itd -p 8081:80 --name firstcontainer --mount type=bind,source=/tmp/mydata,target=/app/data httpd
8d41c7258116686ad8f745e6206154d8c0cd466d9839626378bfb7f2077801cd
[root@ip-172-31-30-82 ec2-user]# docker ps
CONTAINER ID   IMAGE     COMMAND              CREATED             STATUS          PORTS                                      NAMES
8d41c7258116   httpd     "httpd-foreground"   11 seconds ago      Up 10 seconds   0.0.0.0:8081->80/tcp, :::8081->80/tcp      firstcontainer
f8309383c767   httpd     "httpd-foreground"   About an hour ago   Up 6 minutes    0.0.0.0:8080->80/tcp, :::8080->80/tcp      newcontainer
[root@ip-172-31-30-82 ec2-user]#  docker exec -it firstcontainer /bin/bash
root@8d41c7258116:/usr/local/apache2# ls
bin  build  cgi-bin  conf  error  htdocs  icons  include  logs  modules
root@8d41c7258116:/usr/local/apache2# cd /tmp/mydata
bash: cd: /tmp/mydata: No such file or directory
root@8d41c7258116:/usr/local/apache2# cd /app/data
root@8d41c7258116:/app/data# ls
root@8d41c7258116:/app/data# touch file2
root@8d41c7258116:/app/data# ls
file2
```

```
root@8d41c7258116:/app/data# cd /tmp/mydata
bash: cd: /tmp/mydata: No such file or directory
root@8d41c7258116:/app/data# exit
exit
[root@ip-172-31-30-82 ec2-user]# cd /tmp/mydata
[root@ip-172-31-30-82 mydata]# ls
file2
[root@ip-172-31-30-82 mydata]#  docker exec -it firstcontainer /bin/bash
root@8d41c7258116:/usr/local/apache2# cd /tmp/mydata
bash: cd: /tmp/mydata: No such file or directory
root@8d41c7258116:/usr/local/apache2#
```

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Nodejs in Dockerfile

What Node.js actually is

- **JavaScript runtime** built on **V8**
- Runs **outside the browser**
- Designed for **I/O-heavy**, not CPU-heavy, workloads

```
  Running scriptlet: nodejs-1:18.20.8-1.amzn2023.0.2.x86_64                                                                  6/6
INFO: registered node-18 in the alternatives

  Verifying        : libbrotli-1.0.9-4.amzn2023.0.2.x86_64                                                                    1/6
  Verifying        : nodejs-1:18.20.8-1.amzn2023.0.2.x86_64                                                                   2/6
  Verifying        : nodejs-docs-1:18.20.8-1.amzn2023.0.2.noarch                                                              3/6
  Verifying        : nodejs-full-i18n-1:18.20.8-1.amzn2023.0.2.x86_64                                                         4/6
  Verifying        : nodejs-libs-1:18.20.8-1.amzn2023.0.2.x86_64                                                              5/6
  Verifying        : nodejs-npm-1:10.8.2-1.18.20.8.1.amzn2023.0.2.x86_64                                                      6/6

Installed:
  libbrotli-1.0.9-4.amzn2023.0.2.x86_64                        nodejs-1:18.20.8-1.amzn2023.0.2.x86_64                  nodejs-docs-1:18.20.8-1.amzn2023.0.2.noarch
  nodejs-full-i18n-1:18.20.8-1.amzn2023.0.2.x86_64             nodejs-libs-1:18.20.8-1.amzn2023.0.2.x86_64             nodejs-npm-1:10.8.2-1.18.20.8.1.amzn2023.0.2.x86_64

Complete!
[root@ip-172-31-30-82 nodeapp]# npm init -y
Wrote to /home/ec2-user/nodeapp/package.json:

{
  "name": "nodeapp",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "description": ""
}


[root@ip-172-31-30-82 nodeapp]# ls
package.json
```

```
[root@ip-172-31-30-82 nodeapp]#  npm install express

added 65 packages, and audited 66 packages in 4s

22 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
[root@ip-172-31-30-82 nodeapp]# ls
node_modules  package-lock.json  package.json
[root@ip-172-31-30-82 nodeapp]# nano app.js
[root@ip-172-31-30-82 nodeapp]# ls
app.js  node_modules  package-lock.json  package.json
[root@ip-172-31-30-82 nodeapp]# nano Dockerfile
[root@ip-172-31-30-82 nodeapp]# ls
Dockerfile  app.js  node_modules  package-lock.json  package.json
[root@ip-172-31-30-82 nodeapp]# docker build -t node-ec2-app .
[+] Building 22.8s (9/9) FINISHED                                                                                   docker:default
 => [internal] load build definition from Dockerfile                                                                          0.0s
 => => transferring dockerfile: 187B                                                                                          0.0s
 => [internal] load metadata for docker.io/library/node:18                                                                    0.4s
 => [internal] load .dockerignore                                                                                             0.0s
 => => transferring context: 2B                                                                                               0.0s
 => [1/4] FROM docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d96c693a8a4d943498556716d3783            19.8s
```

```
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```

```
 => => sha256:3697be50c98b9d071df4637e1d3491d00e7b9f3a732768c876d82309b3c5a145 1.25MB / 1.25MB                                1.3s
 => => sha256:461077a72fb7fe40d34a37d6a1958c4d16772d0dd77f572ec50a1fdc41a3754d 446B / 446B                                   1.4s
 => => extracting sha256:37927ed901b1b2608b72796c6881bf645480268eca4ac9a37b9219e050bb4d84                                    0.7s
 => => extracting sha256:79b2f47ad44443652b9b5cc81a95ede249fd976310efdbee159f296387837 78c0                                  3.2s
 => => extracting sha256:e23f999113692f62b851cf49a1e93394288a115f5cd2014180e4d3684d34ab                                      8.3s
 => => extracting sha256:cda7f44f2bddcc4bb7514474024b3f3705de00ddb6355a33be5ac7808e5b7125                                    0.0s
 => => extracting sha256:c6b30c3f16966552af10ac00521f60355b1fcfd46ac1c20b1038587e28583ce7                                    2.5s
 => => extracting sha256:3697be50c98b9d071df4637e1d3491d00e7b9f3a732768c876d82309b3c5a145                                    0.1s
 => => extracting sha256:461077a72fb7fe40d34a37d6a1958c4d16772d0dd77f572ec50a1fdc41a3754d                                    0.0s
 => [internal] load build context                                                                                             0.4s
 => => transferring context: 2.19MB                                                                                           0.3s
 => [2/4] WORKDIR /app                                                                                                        0.1s
 => [3/4] COPY . .                                                                                                            0.6s
 => [4/4] RUN npm install                                                                                                     1.6s
 => exporting to image                                                                                                        0.1s
 => => exporting layers                                                                                                       0.1s
 => => writing image sha256:69e19bbafc65d65ddc83090ffaaf635640fb5fa6c7d9b9c17a76174650173493                                  0.0s
 => => naming to docker.io/library/node-ec2-app                                                                               0.0s
[root@ip-172-31-30-82 nodeapp]# docker images
REPOSITORY     TAG       IMAGE ID       CREATED          SIZE
node-ec2-app   latest    69e19bbafc65   55 seconds ago   1.09GB
httpd          latest    14ca0b0b9518   2 weeks ago      117MB
[root@ip-172-31-30-82 nodeapp]# docker run -d -p 3000:3000  node-ec2-app
2d123cc164b281aca4e1ef552459bec69a62fe59c32b1d68ac8edd9143af6c24
[root@ip-172-31-30-82 nodeapp]# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS                                       NAMES
2d123cc164b2   node-ec2-app   "docker-entrypoint.s…"   8 seconds ago    Up 7 seconds    0.0.0.0:3000->3000/tcp, :::3000->3000/tcp   pedantic_herschel
8d41c7258116   httpd          "httpd-foreground"       23 minutes ago   Up 23 minutes   0.0.0.0:8081->80/tcp, :::8081->80/tcp       firstcontainer
f8309383c767   httpd          "httpd-foreground"       2 hours ago      Up 29 minutes   0.0.0.0:8080->80/tcp, :::8080->80/tcp       newcontainer
[root@ip-172-31-30-82 nodeapp]#
```

```
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```

The code inside the **Dockerfile:**

```
[root@ip-172-31-30-82 nodeapp]# cat Dockerfile
FROM node:18

WORKDIR /app

COPY . .

RUN npm install

EXPOSE 3000

CMD ["node", "app.js"]
[root@ip-172-31-30-82 nodeapp]#
```

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The code inside the app.js

```
[root@ip-172-31-30-82 nodeapp]# cat app.js
const express = require('express');
const app = express();

app.get('/', (req, res) => {
  res.send('This is NodeJS running through Docker on EC2 🚀');
});

app.listen(3000, () => {
  console.log('Server running on port 3000');
```

52.87.217.150:3000

Not secure 52.87.217.150:3000

Incognito

This is NodeJS running through Docker on EC2 🚀